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# Deep Learning for Pneumonia Classification from Pediatric Chest X-Rays: From a Binary Baseline to a Bacterial–Viral Three-Class Extension


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**Abstract.** Pneumonia is a leading cause of pediatric mortality worldwide, and chest X-ray (CXR) interpretation remains a bottleneck in resource-limited settings. We study deep-learning models for pneumonia classification on the public Kermany pediatric CXR dataset. As a baseline we address the standard binary task (NORMAL vs. PNEUMONIA), comparing a custom convolutional neural network (CNN) against ImageNet-pretrained ResNet18 and DenseNet121 transfer-learning models. Following feedback to increase the clinical relevance of the task, we extend the problem to a three-class formulation that separates *bacterial* from *viral* pneumonia—a distinction that drives different treatment decisions—using labels derived automatically from the dataset filenames. We further investigate a probability-averaging ensemble, model calibration (Expected Calibration Error and reliability diagrams), decision-threshold optimisation for two clinical scenarios, and Grad-CAM visual explanations that verify the models attend to clinically plausible lung regions. On the binary task the ensemble reaches an accuracy of 0.856, an F1 of 0.896, and an AUROC of 0.963 with 99.7% recall on pneumonia (only 1 of 390 cases missed). On the harder three-class task the ensemble achieves an accuracy of 0.817 and a macro-F1 of 0.804, with per-class F1 scores of 0.77 (NORMAL), 0.91 (BACTERIA), and 0.72 (VIRUS), confirming that viral pneumonia is the most difficult class to recognise. All experiments are reproducible across three seeds and are released as an open, documented code repository.



**Keywords:** Pneumonia classification · Chest X-ray · Transfer learning · Convolutional neural networks · Grad-CAM · Model calibration.


## 1 Introduction

Pneumonia is an acute respiratory infection of the lungs and one of the leading causes of death in children under five years of age worldwide. Chest X-ray (CXR) imaging is the most widely used and accessible modality for diagnosing pneumonia, but its interpretation requires trained radiologists, who are scarce in many low-resource settings. Computer-aided diagnosis systems based on deep learning have therefore attracted strong interest as a way to support, triage, and accelerate radiological workflows.

In this work we develop and evaluate deep-learning models for pneumonia classification from pediatric CXR images, using the public Kermany "Chest X-Ray Images (Pneumonia)" dataset [3]. The project is organised around two tasks. The first is the standard *binary* task of distinguishing NORMAL from PNEUMONIA, which we use as a baseline to compare a custom CNN against two transfer-learning backbones. The second, and our main contribution, is a *three-class* extension that further separates pneumonia into its *bacterial* and *viral* subtypes. This distinction is clinically meaningful: bacterial pneumonia is typically treated with antibiotics, whereas viral pneumonia is not, so a model that can hint at the likely aetiology is potentially more useful in practice than one that only flags the presence of disease.

Our contributions are the following:

1. A clean, reproducible comparison of a custom CNN, ResNet18, and DenseNet121 on binary pediatric pneumonia classification, reported as mean $\pm$ standard deviation across three random seeds.
2. A three-class (NORMAL / BACTERIA / VIRUS) extension whose labels are derived automatically from the dataset filenames, requiring no manual annotation, together with a per-class analysis of where the models succeed and fail.
3. A study of additional, clinically motivated components: a probability-averaging ensemble, a calibration analysis (Expected Calibration Error and reliability diagrams), decision-threshold optimisation for two clinical scenarios, and Grad-CAM visual explanations.
4. A documented, dependency-pinned code repository with command-line entry points that reproduce every result and figure in this report.

This repository is intended for educational experimentation only and is not a clinical decision tool.

## 2 Related Work

The release of large public CXR datasets greatly accelerated research on pneumonia detection. Wang et al. [7] introduced ChestX-ray14, and Rajpurkar et al. [5] proposed CheXNet, a 121-layer DenseNet that matched radiologist performance on pneumonia detection, establishing DenseNet as a strong backbone for CXR analysis. Kermany et al. [3] released the pediatric CXR dataset used in this work and demonstrated the effectiveness of transfer learning from ImageNet-pretrained networks for medical image classification, a strategy that remains a standard and competitive baseline.

Transfer learning is particularly attractive in the medical domain because labelled data is limited: features learned on ImageNet [14] transfer well to CXRs, and only the classifier head (and optionally the deeper layers) needs to be retrained [3,5]. This approach has been applied repeatedly to pneumonia detection on the Kermany dataset with strong results [13]. To make such models trustworthy, two complementary tools are widely used. Grad-CAM [6] produces class-discriminative heatmaps that reveal which image regions a CNN relies on, helping to detect spurious correlations. Calibration analysis, summarised by the Expected Calibration Error and reliability diagrams [8,9], quantifies whether predicted probabilities reflect true correctness likelihood—an important property when a model's output is used to set a decision threshold in a screening or assisted-diagnosis pipeline. Finally, ensembling independently trained models is a simple and reliable way to reduce variance and improve robustness [11], and ensembles of CNNs have been shown to boost pneumonia-detection performance specifically [12].

Most prior work on the Kermany dataset addresses only the binary task. The bacterial-versus-viral distinction is encoded in the filenames but is rarely exploited. Our three-class extension targets exactly this gap and analyses it under the same reproducible protocol as the binary baseline.

## 3 Dataset

We use the Kermany pediatric "Chest X-Ray Images (Pneumonia)" dataset [3,4], which contains anterior–posterior CXRs of pediatric patients (ages 1–5) collected at the Guangzhou Women and Children's Medical Center. The dataset ships with a fixed train/validation/test split and two top-level classes, NORMAL and PNEUMONIA. We use the provided split as-is.

For the three-class task we exploit the fact that every pneumonia image encodes its aetiology in the filename (e.g. `person100_bacteria_4.jpeg` vs. `person154_virus_2.jpeg`). A small parser maps each file to one of NORMAL, BACTERIA, or VIRUS, so the three-class labels are obtained automatically and require no manual annotation. Table 1 summarises the resulting class counts and Fig. 1 visualises the binary split.

Table 1: Image counts for the binary and three-class formulations. The provided validation split contains only 16 images, so for the three-class task a small balanced validation set (8 images per class) is re-sampled deterministically from the training data when classes are missing.

| Split | NORMAL | BACTERIA | VIRUS | Total |
|---|---|---|---|---|
| Train | 1,341 | 2,530 | 1,345 | 5,216 |
| Test | 234 | 242 | 148 | 624 |

Two characteristics of the dataset shape our methodology. First, the training set is *class-imbalanced*: PNEUMONIA images outnumber NORMAL images roughly 2.9 : 1, and within pneumonia, bacterial cases dominate viral cases. Second, the official validation split is tiny (16 images), which makes model selection noisy; for the three-class task we therefore re-sample a small balanced validation

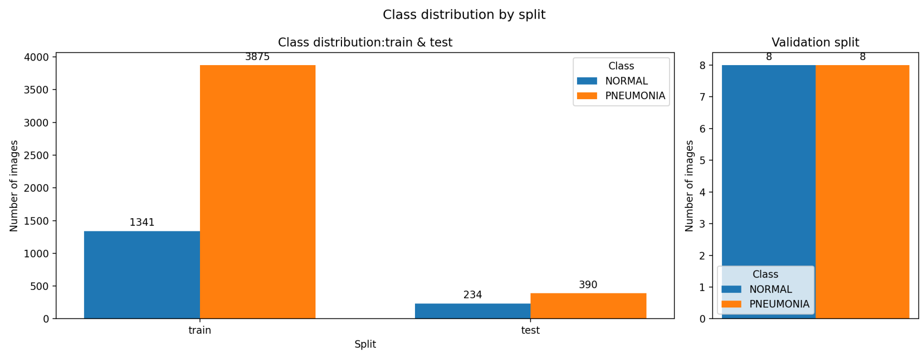



Fig. 1: Binary class distribution by split. *Left:* the training and test sets are dominated by PNEUMONIA ($\approx 2.9\!:\!1$ in training). *Right:* the official validation split contains only 16 images, motivating the re-sampling strategy used for the three-class task.

set from the training data with a fixed seed whenever a class is missing. All reported metrics are computed on the untouched official test set of 624 images.

# 4 Methodology

## 4.1 Preprocessing and Augmentation

All images are resized to $224 \times 224$ pixels. Because the pretrained backbones expect three-channel RGB input, the grayscale X-rays are converted to three channels, and ImageNet normalisation (mean $[0.485, 0.456, 0.406]$, std $[0.229, 0.224, 0.225]$) is applied. During training we use light data augmentation appropriate for radiographs—random horizontal flips and small rotations ($\pm 10°$)—which preserves diagnostic content while improving generalisation. Validation and test images are not augmented.

## 4.2 Models

We compare three architectures.

**Custom CNN (baseline).** A compact three-block convolutional network. Each block is a $3 \times 3$ convolution followed by batch normalisation, a ReLU activation, and $2 \times 2$ max-pooling, with the channel width growing $32 \to 64 \to 128$. A global average-pooling layer, dropout ($p = 0.3$), and a single linear layer produce the output. This model is trained from scratch and serves as a lower-bound reference.

**ResNet18** [1] and **DenseNet121** [2]. Two standard ImageNet-pretrained backbones. We replace the final classification layer with a new linear head sized for the task (one output for binary, three for the three-class task) and apply transfer learning.

### 4.3 Training Protocol

The custom CNN is trained from scratch for up to 20 epochs. The transfer-learning models follow a two-phase schedule. In *phase 1* the backbone is frozen and only the new head is trained for 5 epochs, which adapts the classifier to the new label space without destroying the pretrained features. In *phase 2* the entire network is unfrozen and fine-tuned for up to 30 additional epochs.

All models are optimised with Adam (learning rate $10^{-4}$) and a batch size of 32. The binary task uses binary cross-entropy with logits; the three-class task uses categorical cross-entropy. We use a `ReduceLROnPlateau` scheduler (factor 0.5, patience 3) on the validation loss and early stopping (patience 5). The checkpoint with the lowest validation loss is retained. To assess robustness, every model is trained with three random seeds (0, 1, 2), and results are reported as mean $\pm$ standard deviation. All models are implemented in PyTorch [15], and metrics are computed with scikit-learn [16].

### 4.4 Ensemble

We additionally build a simple ensemble that averages the predicted probabilities of all six transfer-learning checkpoints (ResNet18 and DenseNet121, each with three seeds). For the binary task this averages the sigmoid outputs; for the three-class task it averages the softmax vectors. The averaged probabilities are then thresholded (binary) or `argmax`-ed (three-class) to obtain the final prediction.

### 4.5 Calibration and Threshold Optimisation

Beyond accuracy, we assess whether the predicted probabilities are trustworthy. We compute the Expected Calibration Error (ECE) [8] with ten equal-width confidence bins and plot reliability diagrams (mean accuracy vs. mean confidence per bin). For the binary task we also derive task-specific decision thresholds for two clinical scenarios: a *screening* scenario, in which we pick the highest threshold that still guarantees at least 99% recall on pneumonia (prioritising sensitivity, since missing a sick child is costly), and an *assisted-diagnosis* scenario, in which we pick the threshold that maximises F1 (balancing precision and recall).

### 4.6 Visual Explanations

We use Grad-CAM [6] to produce class-discriminative heatmaps from the last convolutional layer of each transfer-learning model. These overlays let us qualitatively verify that correct predictions are driven by lung-field opacities rather than by irrelevant image artefacts, and they help to interpret failure cases.

### 4.7 Evaluation Metrics

For the binary task we report accuracy, precision, recall (sensitivity), specificity, F1, and AUROC, together with the inference time per image. For the three-class task we report accuracy, macro-averaged precision, recall, and F1, per-class


Fig. 2: Binary DenseNet121 training curves across the three seeds. *Left:* train (dashed) and validation (solid) loss; the train loss converges smoothly while the validation loss is noisy on the small validation set. *Right:* validation F1 saturates near 1.0, showing the validation split is too small to discriminate between checkpoints.

F1, and one-vs-rest macro AUROC. We emphasise recall on the disease classes because, in a clinical context, false negatives are more costly than false positives.

## 5 Results

### 5.1 Training Dynamics

Figure 2 shows the binary DenseNet121 training curves for all three seeds. The two-phase schedule is clearly visible: training loss falls rapidly and smoothly, while validation loss is noisier and plateaus early, after which early stopping halts training (between epochs 11 and 23 depending on the seed). The validation F1 quickly saturates near 1.0 on the tiny 16-image validation split, which confirms that this split is too small for reliable model selection and motivates reporting all final numbers on the full 624-image test set instead. The custom CNN, by contrast, exhibited markedly less stable validation behaviour across seeds, foreshadowing the high variance reported below.

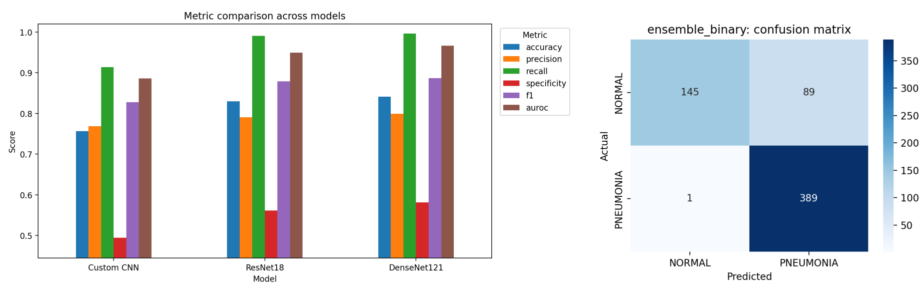
### 5.2 Binary Classification

Table 2 reports the binary results across three seeds, and Fig. 3a visualises the comparison. Both transfer-learning models clearly outperform the custom CNN baseline on every aggregate metric. DenseNet121 is the strongest single model (accuracy 0.841, F1 0.887, AUROC 0.966), narrowly ahead of ResNet18. The custom CNN reaches a competitive F1 but is far less stable: its specificity varies enormously across seeds ($0.494 \pm 0.359$), showing that a from-scratch model on this imbalanced dataset is highly sensitive to initialisation.

The ensemble gives the best overall trade-off: accuracy 0.856, F1 0.896, and a recall of 0.997, meaning it misses only 1 of the 390 pneumonia cases in the test

Table 2: Binary classification (NORMAL vs. PNEUMONIA) on the test set, mean $\pm$ std over seeds $\{0, 1, 2\}$. The ensemble averages the six transfer-learning checkpoints. Best value per column in **bold**.

| Model | Acc. | Prec. | Recall | Spec. | F1 | AUROC |
|---|---|---|---|---|---|---|
| Custom CNN | 0.756 | 0.768 | 0.914 | 0.494 | 0.828 | 0.886 |
| ResNet18 | 0.830 | 0.791 | 0.991 | 0.561 | 0.879 | 0.949 |
| DenseNet121 | 0.841 | 0.799 | **0.997** | **0.581** | 0.887 | **0.966** |
| **Ensemble** | **0.856** | **0.814** | **0.997** | 0.620 | **0.896** | 0.963 |




(a) Per-metric comparison

(b) Ensemble confusion matrix

Fig. 3: Binary results. (a) All models share the same profile: very high recall but comparatively low specificity (red bars). (b) The ensemble misses only 1 of 390 pneumonia cases (FN), at the cost of 89 false positives on NORMAL.

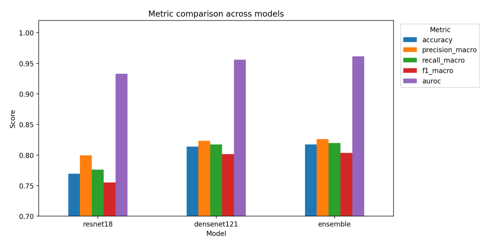
set (confusion matrix in Fig. 3b: TN= 145, FP= 89, FN= 1, TP= 389). The recurring weakness across all models is specificity: the models are eager to predict pneumonia and consequently mislabel a substantial fraction of normal images. This behaviour is the expected and arguably desirable bias for a screening tool, where a missed pneumonia (false negative) is more dangerous than a false alarm, but it does indicate room for improvement on the NORMAL class.

### 5.3 Three-Class Classification

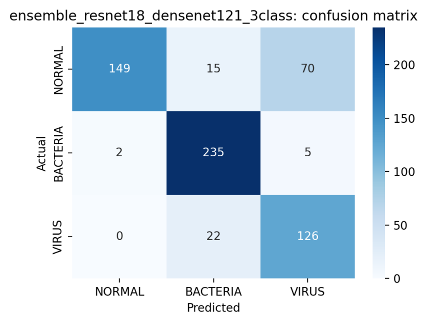
Table 3 reports the three-class results and Fig. 4a visualises them. As expected, separating bacterial from viral pneumonia is substantially harder than the binary task. DenseNet121 again outperforms ResNet18 (accuracy 0.814 vs. 0.769; macro-F1 0.801 vs. 0.755), and the ensemble gives a further small improvement, reaching an accuracy of 0.817, a macro-F1 of 0.804, and a one-vs-rest AUROC of 0.961.

Table 3: Three-class classification (NORMAL / BACTERIA / VIRUS) on the test set, mean ± std over seeds $\{0, 1, 2\}$ (single models) or deterministic for the ensemble. Macro-averaged metrics. Best in **bold**.

| Model | Acc. | Prec.$_M$ | Recall$_M$ | F1$_M$ | AUROC |
|---|---|---|---|---|---|
| ResNet18 | $0.769 \pm 0.005$ | $0.800 \pm 0.011$ | $0.776 \pm 0.008$ | $0.755 \pm 0.002$ | $0.933 \pm 0.016$ |
| DenseNet121 | $0.814 \pm 0.029$ | $0.823 \pm 0.023$ | $0.817 \pm 0.030$ | $0.801 \pm 0.032$ | $0.956 \pm 0.006$ |
| **Ensemble** | **0.817** | **0.826** | **0.820** | **0.804** | **0.961** |




(a) Per-metric comparison




(b) Ensemble confusion matrix

Fig. 4: Three-class results. (a) DenseNet121 and the ensemble dominate ResNet18 on every macro-averaged metric. (b) The dominant error is 70 NORMAL images predicted as VIRUS; BACTERIA is recognised almost perfectly.

### 5.4 Error Analysis

The per-class breakdown of the ensemble is the most informative result. Table 4 reports per-class precision, recall, and F1, and Fig. 4b shows the confusion matrix. Three observations stand out.

Table 4: Per-class metrics of the three-class ensemble on the test set.

| Class | Precision | Recall | F1 | Support |
|---|---|---|---|---|
| NORMAL | 0.987 | 0.637 | 0.774 | 234 |
| BACTERIA | 0.864 | 0.971 | 0.914 | 242 |
| VIRUS | 0.627 | 0.851 | 0.722 | 148 |

First, **bacterial pneumonia is recognised very reliably** (F1 0.914, recall 0.971): it is the majority class and the most visually distinctive, producing dense, lobar consolidation. Second, **viral pneumonia is the hardest class** (F1 0.722): its diffuse, interstitial pattern is easily confused with both normal lungs and early bacterial infiltrates, which is reflected in its low precision (0.627). Third, and most striking, **the dominant error mode is the NORMAL→VIRUS confusion**: 70 of the 234 NORMAL images are predicted as VIRUS, which is precisely why NORMAL recall (0.637) is the lowest of the three classes even though NORMAL precision is the highest (0.987). In other words, the model almost never calls a sick lung normal (only 2 BACTERIA and 0 VIRUS images are misread as NORMAL), but it sometimes reads a faint normal lung as viral pneumonia—a conservative, screening-friendly direction of error. This mirrors the known clinical difficulty of distinguishing subtle viral infiltrates from normal vascular markings.

### 5.5 Inference Cost

All models are lightweight at inference. On our hardware the per-image inference time ranges from roughly 12.6 ms (ResNet18, three-class) to 15.1 ms (DenseNet121, three-class), i.e. all models run at 60–80 images per second on a single GPU. The custom CNN is not meaningfully faster than the transfer-learning models, so its lower accuracy is not compensated by any speed advantage—another reason to prefer the pretrained backbones.

### 5.6 Calibration and Decision Thresholds

Table 5 reports the calibration analysis. The from-scratch custom CNN is, perhaps surprisingly, the best-calibrated binary model (ECE 0.108), whereas the transfer-learning models are overconfident (ECE 0.153–0.160): they tend to push probabilities towards the extremes. The three-class models are less well calibrated still (ECE 0.245–0.276), reflecting the greater difficulty of the task. Figure 5 shows the corresponding reliability diagrams; the transfer-learning models concentrate their predictions in the high-confidence bins, far from the diagonal.

Table 5: Expected Calibration Error (ECE, 10 bins, lower is better) and optimised binary decision thresholds. "Screening" is the highest threshold maintaining $\geq 99\%$ recall; "F1" maximises F1. Thresholds are only defined for the binary task.

| Model | ECE | Thr. (screening) | Thr. (F1) |
|---|---|---|---|
| Custom CNN | $0.108 \pm 0.051$ | 0.245 | 0.596 |
| ResNet18 | $0.160 \pm 0.016$ | 0.593 | 0.981 |
| DenseNet121 | $0.153 \pm 0.015$ | 0.860 | 0.995 |
| Ensemble (binary) | 0.156 | 0.751 | 0.986 |
| ResNet18 (3-class) | $0.266 \pm 0.086$ | – | – |
| DenseNet121 (3-class) | $0.245 \pm 0.027$ | – | – |
| Ensemble (3-class) | 0.276 | – | – |

The threshold analysis quantifies an important operational point. Because the transfer-learning models are confident, the default 0.5 threshold is not opti-

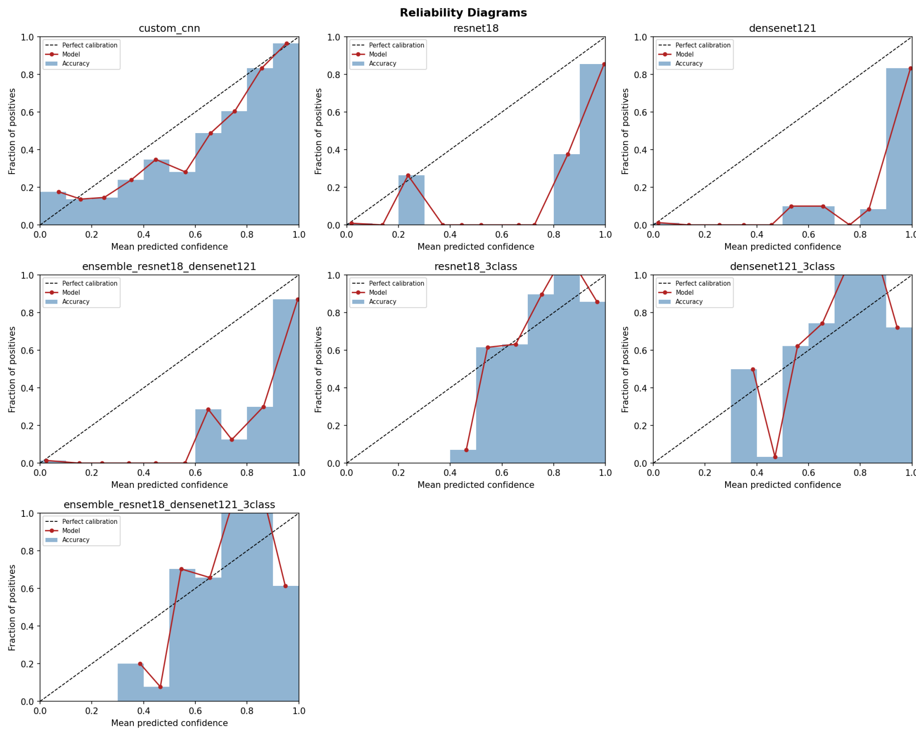
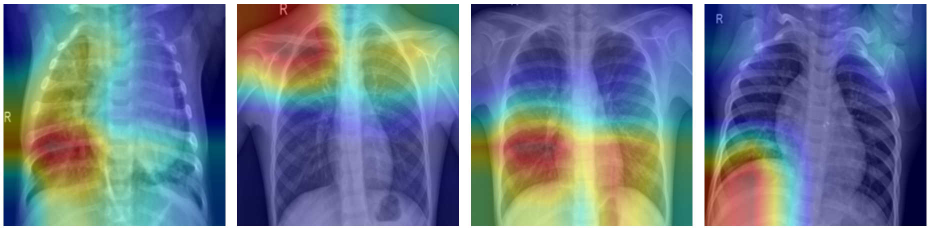



Fig. 5: Reliability diagrams. The dashed diagonal is perfect calibration; points above the diagonal indicate under-confidence and below it over-confidence. Transfer-learning models concentrate their predictions in the high-confidence bins.

mal: maximising F1 requires very high thresholds (e.g. 0.995 for DenseNet121), whereas a screening deployment that demands $\geq 99\%$ recall can afford a much lower threshold. Exposing these two operating points makes the same trained model usable in both a high-sensitivity triage mode and a balanced assisted-diagnosis mode, without retraining.

### 5.7 Grad-CAM Visual Explanations

Figure 6 shows binary Grad-CAM overlays for DenseNet121 across the four outcome types. For true positives and true negatives the activation is anatomically sensible—concentrated on the consolidated lung field for pneumonia and diffuse/peripheral for normal lungs—whereas the false positive and false negative reveal where the model is led astray. Figure 7 extends this to the three-class model: correct bacterial and viral predictions focus on the affected lung fields, while the NORMAL→VIRUS failure shows diffuse activation over otherwise normal lungs, visually corroborating the dominant error mode identified in Table 4.



(a) TP (pneumonia) (b) TN (normal) (c) FP (d) FN

Fig. 6: Binary Grad-CAM overlays (DenseNet121) for the four outcome types. Correct predictions (TP, TN) attend to plausible lung regions; the FP and FN illustrate the failure modes behind the imperfect specificity.

(a) Correct: BACTERIA (b) Correct: VIRUS (c) Error: NORMAL as VIRUS

Fig. 7: Three-class Grad-CAM overlays (DenseNet121). Correct predictions focus on the affected lung fields; the failure case shows diffuse activation over normal lungs.

## 6 Discussion

Three findings stand out. First, transfer learning is decisively better than training from scratch on this dataset: ResNet18 and DenseNet121 outperform the custom CNN on every metric and, crucially, are far more stable across seeds. Given limited medical data, ImageNet-pretrained features are a strong and cheap prior. Second, the three-class extension is both feasible and informative. Although accuracy drops relative to the binary task (as expected), the model still separates bacterial from viral pneumonia well above chance, and the per-class analysis localises the difficulty precisely at the NORMAL/VIRUS boundary, which mirrors the known clinical difficulty of distinguishing viral infiltrates from normal variation. Third, the auxiliary analyses—ensembling, calibration, and threshold optimisation—turn a raw classifier into something closer to a deployable component, by reducing variance, exposing over-confidence, and providing explicit operating points for screening versus assisted diagnosis.

*Limitations.* Several caveats apply. (i) The dataset comes from a single centre and a narrow pediatric age range, so the models may not generalise to other populations or acquisition protocols. (ii) The validation split is tiny, making model selection noisy; we mitigated but did not eliminate this with re-sampling. (iii) The bacterial/viral labels are derived from filenames and inherit the dataset's original (possibly imperfect) labelling. (iv) We did not explicitly correct for class imbalance with re-weighting or re-sampling beyond augmentation, which partly explains the low specificity. (v) All metrics are on a single test set; external validation would be needed before any real-world claim. We reiterate that the system is for educational use only and is not a diagnostic device.

## 7 Conclusion and Future Work

We presented a reproducible study of deep-learning models for pediatric pneumonia classification, progressing from a binary baseline to a clinically motivated three-class extension that distinguishes bacterial from viral pneumonia. Transfer-learning backbones, and especially a simple ensemble of ResNet18 and DenseNet121, gave the best results: an F1 of 0.896 and 99.7% pneumonia recall on the binary task, and a macro-F1 of 0.804 on the three-class task. Calibration, threshold optimisation, and Grad-CAM analyses added interpretability and operational flexibility.

Future work includes addressing class imbalance with weighted losses or focal loss [10], temperature scaling [8] to improve calibration, stronger backbones and test-time augmentation, and—most importantly—external validation on independent CXR datasets to test generalisation beyond the single source used here. All code, configurations, and scripts needed to reproduce every number and figure in this report are released in the accompanying repository.

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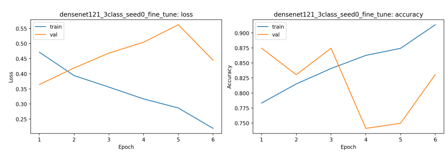
# A Appendix: Per-Seed Training and Evaluation

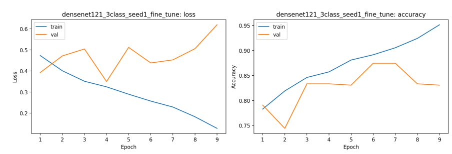
This appendix provides the full per-seed material that supports the aggregated results in the main text. It is supplementary and does not count towards the page limit.

## A.1 Three-Class Training Curves (per seed)

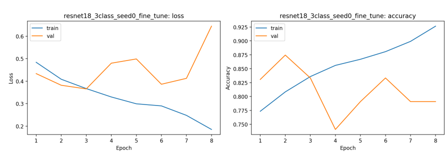
Figure 8 shows the fine-tuning loss and validation-F1 curves for both backbones across the three seeds. DenseNet121 converges to a lower and more stable validation loss than ResNet18, consistent with its higher final accuracy.



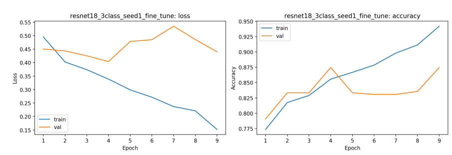

(a) DenseNet121, seed 0




(b) DenseNet121, seed 1




(c) ResNet18, seed 0

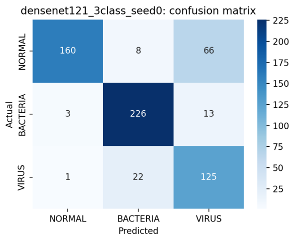



(d) ResNet18, seed 1

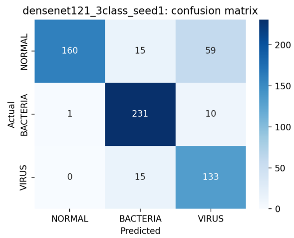
Fig. 8: Three-class fine-tuning curves (representative seeds) for both backbones.

## A.2 Three-Class Confusion Matrices (per seed)

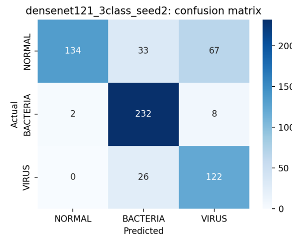
Figure 9 shows the per-seed confusion matrices for DenseNet121. The NORMAL→VIRUS confusion is consistent across seeds, confirming it is a systematic property of the task rather than an artefact of a single run.




(a) seed 0




(b) seed 1



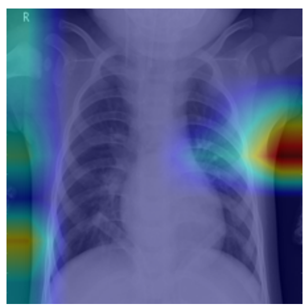

(c) seed 2

Fig. 9: Per-seed three-class confusion matrices for DenseNet121.

### A.3 Additional Grad-CAM Examples

Figure 10 provides further three-class Grad-CAM examples, including additional failure modes (BACTERIA misread as VIRUS, and NORMAL misread as BACTERIA).


(a) Correct: NORMAL




(b) Error: BACTERIA as VIRUS

(c) Error: NORMAL as BACTERIA

Fig. 10: Additional three-class Grad-CAM overlays (DenseNet121).